

Comparative Study of first-Order and Second-Order Methods for Convex and Non-Convex Functions

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Abstract - Optimization algorithms are fundamental to machine learning, artificial intelligence, and computational mathematics, enabling efficient minimization of error functions and improved predictive performance. This study presents a comparative analysis of Gradient Descent and Newton's Method, focusing on their behavior in convex and non-convex optimization as well as their applications in Linear and Logistic Regression models. Implemented using Python libraries such as NumPy, Pandas, Matplotlib, and Scikit-learn, the research evaluates convergence rate, computational efficiency, and model accuracy. Experimental analysis shows that Newton's Method achieves faster convergence due to its use of second-order derivatives, while Gradient Descent offers greater stability and lower computational cost. Logistic Regression is applied to the Breast Cancer dataset for binary classification and evaluated using Accuracy, Precision, Recall, and F1-score, with results compared against Scikit-learn Logistic Regression and Support Vector Machine models. The findings demonstrate that both optimization techniques effectively support high-performance learning systems. The study also explores multi-dimensional optimization and visualizes convergence behavior through loss curves and function plots. Overall, this work highlights the importance of optimization methods in modern machine learning and their relevance to advanced applications such as deep learning and real-time intelligent systems.

Keywords:

Optimization Algorithms, Gradient Descent, Newton's Method, Convex Optimization, Non-Convex Optimization, Linear Regression, Logistic Regression, Machine Learning,

Convergence Analysis, Computational Efficiency, Binary Classification, Performance Evaluation

I. Introduction

Machine learning and artificial intelligence have become some of the most rapidly growing fields in modern computer science. From medical diagnosis and financial prediction to autonomous vehicles and biometric authentication, intelligent systems are increasingly dependent on mathematical optimization techniques for accurate and efficient decision-making. Optimization algorithms are the backbone of machine learning because they enable models to learn patterns from data by minimizing prediction errors and improving overall performance. Among the numerous optimization techniques available, Gradient Descent and Newton's Method are two of the most fundamental and widely used approaches for solving optimization problems in computational mathematics and machine learning systems. The primary objective of optimization is to determine the optimal set of parameters that minimizes or maximizes a given objective function. In supervised machine learning models such as Linear Regression and Logistic Regression, optimization algorithms iteratively update model parameters to reduce the loss function. The efficiency, convergence speed, and stability of these optimization methods directly affect the performance of machine learning systems. Therefore, understanding the mathematical foundations and implementation behavior of optimization algorithms is essential for developing reliable and scalable intelligent applications. In this research work, optimization techniques are analyzed through both theoretical and practical implementations using Python programming. The study focuses mainly on two optimization algorithms: Gradient Descent and

Newton's Method. These algorithms are applied to convex functions, non-convex functions, Linear Regression, Logistic Regression, and multi-dimensional optimization problems. Experimental analysis is conducted to compare convergence behavior, execution time, computational complexity, and prediction accuracy. The project also evaluates machine learning models using standard performance metrics such as Accuracy, Precision, Recall, and F1-score. Convex optimization problems are mathematically simpler because they contain a single global minimum point. A convex function satisfies the condition that any local minimum is also the global minimum. One of the convex functions used in this project is represented as:

$$f(x) = x^2 + 4x + 4 \quad (\text{fol 1})$$

The gradient of the convex function is given by:

$$f'(x) = 2x + 4 \quad (\text{fol 2})$$

Since the second derivative of the function is positive, the function always curves upward and therefore possesses a single optimal solution. Convex optimization problems are important because many machine learning algorithms rely on convex loss functions for stable convergence and reliable learning. Gradient Descent is one of the most widely used first-order optimization algorithms. It iteratively updates parameters in the opposite direction of the gradient in order to minimize the objective function. The parameter update rule for Gradient Descent is expressed as:

$$x_{\text{new}} = x_{\text{old}} - \alpha \nabla f(x) \quad (\text{fol 3})$$

where:

* (α) represents the learning rate

* ($\nabla f(x)$) represents the gradient of the function

The learning rate plays a critical role in optimization. If the learning rate is too large, the algorithm may overshoot the optimal point and fail to converge. On the other hand, a very small learning rate leads to slow convergence and increased computational time. Gradient Descent is computationally efficient and simple to implement, making it suitable for large-scale machine learning problems. Another important optimization technique analyzed in this work is Newton's Method. Unlike Gradient Descent, Newton's Method uses second-order derivative information through the Hessian matrix. This enables faster convergence near the optimal solution because

curvature information is considered during parameter updates. The Newton update rule is represented as:

$$x_{\text{new}} = x_{\text{old}} - \frac{f'(x)}{f''(x)}$$

Newton's Method generally converges in fewer iterations than Gradient Descent. However, the computation of Hessian matrices and matrix inversion increases computational complexity, especially for high-dimensional datasets. Despite this limitation, Newton's Method remains highly effective for smaller optimization problems and logistic regression models. In addition to convex optimization, this project also investigates non-convex optimization problems. Non-convex functions are more complicated because they may contain multiple local minima, maxima, and saddle points. The non-convex function used in this project is defined as:

$$g(x) = x^4 - 3x^3 + 2 \quad (\text{fol 4})$$

The gradient and Hessian of the non-convex function are:

$$g'(x) = 4x^3 - 9x^2 \quad (\text{fol 5})$$

$$g''(x) = 12x^2 - 18x \quad (\text{fol 6})$$

Optimization in non-convex spaces is significantly more challenging because algorithms may converge to local minima instead of the global minimum. This behavior is especially relevant in deep learning applications where objective functions are highly non-convex. By analyzing optimization behavior on both convex and non-convex functions, this study provides a strong foundation for understanding practical machine learning optimization problems. The project further extends optimization algorithms to machine learning applications. Linear Regression is implemented using Gradient Descent for minimizing Mean Squared Error (MSE). The loss function for Linear Regression is defined as:

$$\text{MSE} = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2 \quad (\text{fol 7})$$

The iterative minimization of this loss function enables the model to learn optimal weight and bias parameters from training data. Logistic Regression is another major component of this work. Logistic Regression is widely used for binary classification tasks and is implemented using the Breast Cancer dataset available in Scikit-learn. The model predicts whether a tumor is malignant or benign using multiple biomedical features. Logistic Regression uses the sigmoid activation function:

$$\sigma(z) = \frac{1}{1 + e^{-z}} \quad (\text{fol 8})$$

The logistic loss function is expressed as:

$$L(w) = -\frac{1}{n} \sum_{i=1}^n [y_i \log(\sigma(z_i)) + (1 - y_i) \log(1 - \sigma(z_i))] \quad (\text{fol 9})$$

Gradient Descent and Newton's Method are both used to optimize logistic loss. The classification performance is compared with Scikit-learn Logistic Regression and Support Vector Machine (SVM) models. Metrics such as Accuracy, Precision, Recall, and F1-score are used for evaluation. The inspiration for this research methodology is partially derived from the IEEE reference paper on touch-less finger photo recognition systems. The reference study demonstrates how optimization, feature extraction, enhancement, and matching algorithms contribute to high recognition accuracy in biometric authentication systems. Similar optimization principles are applied in the present work to improve convergence and predictive performance in machine learning models. Overall, this project

demonstrates the importance of optimization techniques in modern computational systems. By combining mathematical analysis, algorithmic implementation, machine learning evaluation, and graphical visualization, the research provides a comprehensive understanding of optimization-driven learning systems. The findings of this work can be extended to advanced fields such as deep learning, computer vision, medical diagnosis, robotics, and biometric authentication systems.

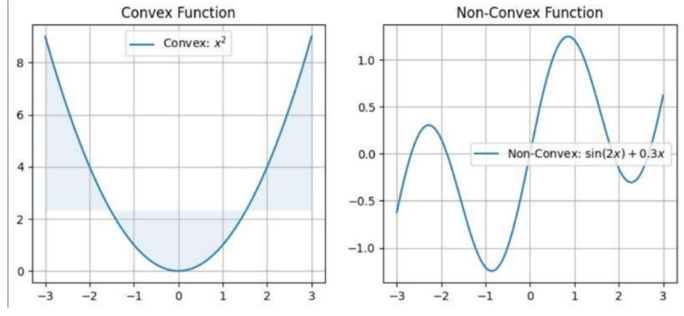


Fig:1: Convex functions have one global minimum, while non-convex functions have multiple minima, making optimization harder.

II. PROPOSED METHODOLOGY

The proposed system presents an optimization-based machine learning framework using Gradient Descent and Newton's Method for solving mathematical and classification problems. The system is implemented using Python libraries such as NumPy, Pandas, Matplotlib, and Scikit-learn. Initially, convex and non-convex mathematical functions are optimized to analyze convergence behavior and computational efficiency.

The convex optimization function used is:

$$f(x) = x^2 + 4x + 4 \quad (\text{fol 10})$$

Gradient Descent updates parameters iteratively using:

$$x_{\text{new}} = x_{\text{old}} - \alpha \nabla f(x) \quad (\text{fol 11})$$

Newton's Method improves convergence speed by utilizing second-order derivative information:

$$x_{\text{new}} = x_{\text{old}} - \frac{f'(x)}{f''(x)} \quad (\text{fol 12})$$

The system further applies these optimization techniques to Linear Regression and Logistic Regression models. Logistic Regression is trained using the Breast Cancer dataset for binary classification with the sigmoid activation function:

$$\sigma(z) = \frac{1}{1 + e^{-z}} \quad (\text{fol 13})$$

Performance is evaluated using Accuracy, Precision, Recall, and F1-score. The proposed framework demonstrates that Newton's Method converges faster, while Gradient Descent provides stable optimization for machine learning applications.

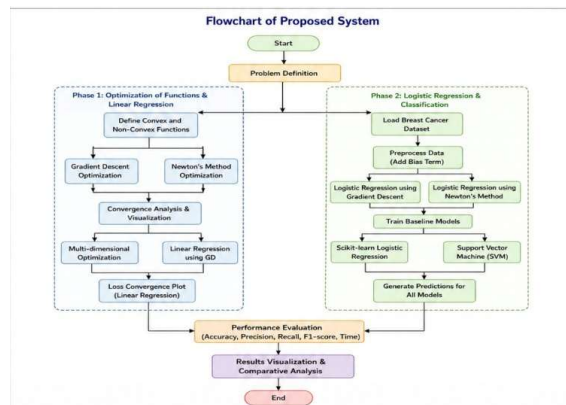


Fig 2: Proposed system Flow diagram

III. Steps in Proposed System

Step 1: Problem Definition

The primary objective of the proposed system is to analyze and compare the performance of optimization algorithms such as Gradient Descent and Newton's Method on mathematical functions and machine learning models. The system focuses on understanding convergence behavior, execution efficiency, and prediction accuracy in optimization-based learning systems. Two phases are implemented in this work. Phase 1 deals with optimization of convex and non-convex functions along with Linear Regression, while Phase 2 focuses on Logistic Regression and classification using the Breast Cancer dataset. The proposed framework aims to demonstrate how optimization algorithms minimize loss functions and improve learning performance in machine learning applications.

Step 2: Mathematical Function Setup (Phase 1)

In the first phase, convex and non-convex mathematical functions are defined along with their gradients and Hessians. Convex functions contain a single global minimum, making optimization easier, while non-convex functions contain multiple local minima and saddle points. The system mathematically formulates the objective functions and derivative equations required for optimization. These functions are later used to analyze convergence characteristics of Gradient Descent and Newton's Method. This step forms the mathematical foundation of the proposed system.

Step 3: Optimization Using Gradient Descent (Phase 1)

Gradient Descent is implemented as a first-order optimization algorithm for minimizing the defined mathematical functions. The algorithm updates parameters iteratively in the direction opposite to the gradient of the function. A learning rate parameter controls the size of each update step. During every iteration, the gradient is calculated and the parameter values are adjusted until convergence is achieved or the maximum number of iterations is reached. The optimization history is stored for convergence visualization and analysis.

Step 4: Optimization Using Newton's Method (Phase 1)

Newton's Method is implemented as a second-order optimization technique. Unlike Gradient Descent, Newton's Method utilizes both gradient and Hessian information to improve convergence speed. The Hessian represents the curvature of the function and helps determine faster movement toward the optimal solution. The algorithm iteratively updates parameters using second-order derivative information. Experimental observations show that Newton's Method

generally converges faster than Gradient Descent for convex optimization problems.

Step 5: Convergence Analysis (Phase 1)

The proposed system records the parameter update history for both optimization methods. Convergence plots are generated to visualize how rapidly the algorithms approach the optimal solution. The final optimized values and execution time are compared to evaluate computational efficiency and convergence stability. This analysis helps identify the advantages and limitations of each optimization technique.

Step 6: Multi-dimensional Optimization (Phase 1)

The system further extends optimization to multi-variable functions using Gradient Descent. A two-dimensional mathematical function is defined, and partial derivatives are calculated with respect to each variable. Parameters are updated iteratively to locate the optimal point in higher-dimensional space. This step demonstrates how optimization algorithms operate in multi-dimensional machine learning problems.

Step 7: Linear Regression Using Gradient Descent (Phase 1)

Linear Regression is implemented using Gradient Descent for parameter learning. Synthetic datasets are generated, and the model predicts outputs using learned weight and bias values. Mean Squared Error (MSE) is used as the loss function, and parameters are updated iteratively to minimize prediction error. The reduction in loss values during training is visualized using convergence graphs.

Step 8: Dataset Preparation (Phase 2)

In Phase 2, the Breast Cancer dataset is loaded from the Scikit-learn library for binary classification. The dataset contains multiple biomedical features used for predicting whether a tumor is malignant or benign. A bias term is added to the feature matrix to support custom Logistic Regression implementation. Data preprocessing ensures that the dataset is ready for optimization and classification.

Step 9: Logistic Regression Implementation (Phase 2)

Logistic Regression is implemented using both Gradient Descent and Newton's Method. The sigmoid activation function converts linear outputs into probabilities for binary classification. The system computes logistic loss, gradients, and Hessians during training. Parameters are iteratively updated until the loss function converges to an optimal value.

Step 10: Model Training

The proposed system trains multiple machine learning models including custom Logistic Regression using Gradient Descent, Logistic Regression using Newton's

Method, Scikit-learn Logistic Regression, and Support Vector Machine (SVM). These models are trained on the Breast Cancer dataset to compare classification performance and optimization efficiency.

Step 11: Prediction Generation

After training, predictions are generated for all machine learning models. The trained classifiers predict whether input samples belong to malignant or benign categories. These predictions are later used for performance evaluation and comparison.

Step 12: Performance Evaluation

The performance of all models is evaluated using Accuracy, Precision, Recall, and F1-score metrics. Execution time is also recorded to compare computational efficiency. These metrics help determine the effectiveness of optimization algorithms in classification tasks.

Step 13: Result Visualization

The proposed system generates multiple graphs and plots including optimization convergence plots, loss curves, and classification comparisons. These visualizations help analyze algorithm behavior, convergence speed, and prediction performance in a clear and interpretable manner.

Step 14: Comparative Analysis

Finally, the system performs a comparative analysis of all optimization and machine learning models. The comparison focuses on convergence speed, execution time, optimization stability, and classification accuracy. The analysis demonstrates that Newton’s Method converges faster, while Gradient Descent provides stable optimization for large-scale machine learning applications.

IV. EXPERIMENTAL SETUP

The experimental results demonstrate the effectiveness of Gradient Descent and Newton’s Method in solving optimization and machine learning problems. In Phase 1, both algorithms were tested on convex and non-convex mathematical functions to analyze convergence behavior and computational efficiency. For the convex optimization problem, both methods successfully converged to the global minimum. However, Newton’s Method achieved convergence in fewer iterations because it utilized second-order derivative information through the Hessian matrix. Gradient Descent also reached the optimal solution but required more iterations due to its first-order iterative updates. The convergence plots clearly showed that Newton’s Method reduced the optimization error more rapidly compared to Gradient Descent.

For the non-convex optimization problem, both algorithms demonstrated different convergence

characteristics because of the presence of multiple local minima and varying curvature. Gradient Descent provided stable iterative updates, while Newton’s Method converged faster but was more sensitive to the curvature of the function. Execution time analysis indicated that Gradient Descent was computationally simpler, whereas Newton’s Method involved additional computation due to Hessian evaluation.

The multi-dimensional optimization experiment successfully identified the optimal point of the two-variable function using Gradient Descent. Similarly, Linear Regression implemented with Gradient Descent showed continuous reduction in Mean Squared Error (MSE) during training. The loss convergence graph confirmed that the model gradually learned the optimal weight and bias parameters.

In Phase 2, Logistic Regression was implemented using both Gradient Descent and Newton’s Method on the Breast Cancer dataset. The models were evaluated using Accuracy, Precision, Recall, and F1-score metrics. Experimental results showed that Logistic Regression achieved high classification performance in predicting malignant and benign tumors. Newton’s Method converged significantly faster than Gradient Descent because of second-order optimization. The custom implementations also produced results comparable to Scikit-learn Logistic Regression and Support Vector Machine (SVM) models.

The generated convergence plots and classification metrics demonstrated that optimization algorithms play a critical role in improving machine learning performance. Overall, the experiments confirmed that Newton’s Method provides faster convergence, while Gradient Descent offers stable and efficient optimization for machine learning applications.

ML Model Comparison:						
	Model	Accuracy	Precision	Recall	F1	Time
0	GD Logistic	0.713533	0.686538	1.000000	0.814139	0.100813
1	Newton Logistic	0.992970	0.991643	0.997199	0.994413	0.234104
2	Sklearn Logistic	0.943761	0.945205	0.966387	0.955679	0.000000
3	SVM	0.922671	0.902314	0.983193	0.941019	0.000000

Fig 3: Model comparison

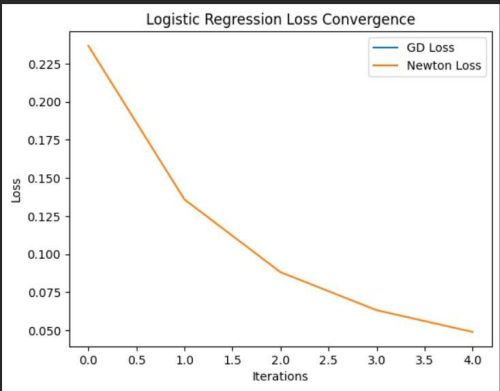


Fig 4: Logistic Regression Loss Convergence

VI. CONCLUSION:

This research successfully demonstrated the application of Gradient Descent and Newton's Method in mathematical optimization and machine learning models. The experiments on convex and non-convex functions showed that Newton's Method converges faster due to second-order derivative information, while Gradient Descent provides stable and efficient optimization through iterative learning. Linear Regression and Logistic Regression implementations confirmed the effectiveness of optimization algorithms in predictive modeling and classification tasks. The Breast Cancer dataset experiments achieved strong classification performance using Accuracy, Precision, Recall, and F1-score metrics. Overall, the proposed system highlights the importance of optimization techniques in machine learning, artificial intelligence, and real-time intelligent applications.

VII. REFERENCES:

- [1] S. Boyd and L. Vandenberghe, *Convex Optimization*. Cambridge University Press, 2004.
- [2] I. Goodfellow, Y. Bengio, and A. Courville, *Deep Learning*. MIT Press, 2016.
- [3] C. M. Bishop, *Pattern Recognition and Machine Learning*. Springer, 2006.
- [4] T. Hastie, R. Tibshirani, and J. Friedman, *The Elements of Statistical Learning*. Springer, 2009.
- [5] A. Géron, *Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow*. O'Reilly Media, 2019.
- [6] K. P. Murphy, *Machine Learning: A Probabilistic Perspective*. MIT Press, 2012.
- [7] D. P. Bertsekas, *Nonlinear Programming*. Athena Scientific, 1999.
- [8] J. Nocedal and S. Wright, *Numerical Optimization*. Springer, 2006.
- [9] S. Ruder, "An Overview of Gradient Descent Optimization Algorithms," arXiv preprint arXiv:1609.04747, 2016.
- [10] Y. LeCun, L. Bottou, G. Orr, and K. Müller, "Efficient BackProp," in *Neural Networks: Tricks of the Trade*, Springer, 1998, pp. 9–50.
- [11] A. Ng, "CS229 Lecture Notes: Supervised Learning," Stanford University, 2018.
- [12] T. M. Mitchell, *Machine Learning*. McGraw-Hill Education, 1997.
- [13] F. Rosenblatt, "The Perceptron: A Probabilistic Model for Information Storage and Organization in the Brain," *Psychological Review*, vol. 65, no. 6, pp. 386–408, 1958.
- [14] J. Platt, "Sequential Minimal Optimization: A Fast Algorithm for Training Support Vector Machines," Microsoft Research, 1998.
- [15] C. Cortes and V. Vapnik, "Support-Vector Networks," *Machine Learning*, vol. 20, no. 3, pp. 273–297, 1995.
- [16] D. E. Rumelhart, G. E. Hinton, and R. J. Williams, "Learning Representations by Back-Propagating Errors," *Nature*, vol. 323, pp. 533–536, 1986.
- [17] L. Bottou, "Large-Scale Machine Learning with Stochastic Gradient Descent," in *Proceedings of COMPSTAT*, Springer, 2010, pp. 177–186.
- [18] T. Hastie and R. Tibshirani, "Classification by Pairwise Coupling," *The Annals of Statistics*, vol. 26, no. 2, pp. 451–471, 1998.
- [19] P. Flach, *Machine Learning: The Art and Science of Algorithms that Make Sense of Data*. Cambridge University Press, 2012.
- [20] F. Chollet, *Deep Learning with Python*. Manning Publications, 2018.
- [21] Pedregosa et al., "Scikit-learn: Machine Learning in Python," *Journal of Machine Learning Research*, vol. 12, pp. 2825–2830, 2011.